Labor Markets Check-in 2: Half PAP

Research question: Does employer callback behavior toward women differ between remote and in-person jobs in the U.S. tech sector?

Group Members: Maimoona, Stephanie, Zheng, Ziqiao

Introduction

Despite decades of progress in education and labor market participation, women remain underrepresented in science, technology, engineering, and mathematics (STEM) occupations in the United States. This gender gap is especially pronounced in the tech sector, where women continue to face barriers to entry, retention, and advancement. Much existing research has sought to explain this disparity, pointing to factors such as differences in educational pathways, workplace culture, implicit bias, and employer discrimination. Among these, hiring discrimination remains a central concern, as it directly shapes access to opportunities at the earliest stage of employment.

Numerous existing studies, particularly those investigating racial discrimination in the labor market, have found that callback rates after resumes are sent are a useful metric for measuring discrimination. However, much of this research has focused on traditional, in-person work environments. With the rapid expansion of remote work, accelerated by the COVID-19 pandemic, the hiring process itself may be evolving in ways that differentially affect gender bias. Remote work may reduce certain forms of discrimination by limiting face-to-face interaction, or alternatively, it may exacerbate bias if employers rely more heavily on stereotypes in the absence of personal contact.

This paper seeks to address a gap in the literature by examining whether employer callback behavior toward women differs between remote and in-person job postings in the U.S. tech sector. Specifically, we ask: Does the likelihood of receiving a callback vary by applicant gender depending on whether the position is remote or in-person? To answer this question, we design an experiment in which fictitious resumes, equivalent in qualifications but differing in gender signals, are submitted to job postings across both work modalities. By comparing callback rates across these conditions, we aim to isolate the role of job format in shaping gender discrimination at the hiring stage.

Understanding whether and how remote work influences hiring bias has important implications for both policy and organizational practice. If remote work reduces discriminatory behavior, expanding access to such opportunities could help promote gender equity in tech. Conversely, if bias persists or intensifies in remote hiring contexts, targeted interventions may be necessary to ensure fair and inclusive recruitment processes.

Experimental Design

It is unclear if the underrepresentation of women in the tech workforce is reflected in alternative working arrangements, such as hybrid or remote work (U.S. Equal Employment Opportunity Commission, 2024). The 2020 COVID-19 pandemic boosted remote work as many industries adopted a hybrid or fully remote policy, many of which remained in place even after social distancing policies were scaled back. Between 2019 and 2021, remote work increased significantly across all major industry groups, with the largest growth occurring in the professional, scientific, and technical services industry (Pabilonia and Redmond 2024). Although the share of remote workers in professional, scientific, and technical services occupations decreased slightly between 2021 and 2022, it remains higher than its 2019 value, demonstrating the enduring nature of remote work (ibid).

Since the ubiquitousness of remote work is still relatively nascent, little is known about whether the gender gap is as pervasive as in traditional in-person working arrangements—or if there is a gender gap at all. Often, gender discrimination manifests during the hiring stage; in fact, survey data indicates that tech recruiters even acknowledge that bias exists against female candidates (WomenTech Network 2026). Studying callback rates for male and female applicants in remote tech jobs and comparing those callback rates to those of male and female applicants in in-person tech jobs can offer insights into whether certain work environments are more conducive to reducing the gender gap and increasing the share of women in the tech workforce.

Our study will randomly vary an applicant's gender and the type of work environment they are applying for in order to isolate employer behavior for both male and female candidates and for both remote and in-person jobs. For each unique application, we will track the callback rate (the rate at which a candidate for a certain job is contacted about their application or invited to participate in an interview). We will compare the callback rates between male and female applicants in both remote and in-person jobs in the professional, scientific, and technical services industry in the United States.

We will employ an audit study in which resumes are randomly assigned to be sent to either in-person job postings or remote job postings. Our study participants are companies in the professional, scientific, and technical services industry. As previously discussed, this industry experienced the biggest boost in remote work during and after the COVID-19 pandemic. It is also a particularly male-dominated industry where women make up just over a quarter of the workforce (WomenTech Network 2026). We will limit our sample to employers based in Virginia, which has the largest number of job postings in the tech industry across the United States (Robinson 2026). Limiting our sample population to companies specifically within this region will help isolate the causal relationship between our intervention and outcome variables and reduce potential spurious variables that may influence the outcome variable, such as

workforce shortages that stimulate labor supply and demand, the job market of a particular state, and the share of women and men in the labor force. However, the differences in labor supply and demand across different states and municipalities may pose threats to external validity (for example, tech companies in states with greater demand for tech workers may be less likely to discriminate against women in the hiring process due to workforce needs, so our results may not be generalizable across the entire United States. Further, since Virginia has the largest number of tech jobs in the United States, we might expect a higher share of female candidates).

We will vary both gender and work environment (in-person or remote) in our study. Fictitious applications will be randomly assigned to a gender signal (male or female) and randomly sent to either an in-person or remote job. Each application will only be sent to one job posting. We will be comparing the callback rates for in-person and remote jobs for male and female applicants to assess whether remote work is associated with less hiring bias in the form of increased callback rates.

It is possible we may face attrition in our study, which may differ across treatment and control groups. For example, a job posting might be taken down without hiring a candidate. We do not anticipate spillovers: each application will be randomly assigned to a gender signal and randomly sent to a job posting that is remote or in-person. Applicants will only send their resume to in-person jobs or remote jobs and not both. Therefore, randomization is the correct approach for this study. A resume cannot have both gender signals, nor can it be sent to both a remote and an in-person job posting.

Data

The key inputs in our design are the gender signal of the applicant, which is randomized through the name on the resume, and the work environment of the job, categorized as either remote or in-person. All other resume characteristics, such as education, experience, and skills, are held constant to ensure comparability across applications. The main output of interest is the employer's callback behavior, measured as a binary indicator equal to one if the applicant receives a positive response (such as an interview invitation or request for next steps) and zero otherwise. The primary outcome is the difference in callback rates between male and female applicants and how this gap varies across remote and in-person job settings.

We will collect data through an audit experiment in which standardized resumes are submitted to real job postings in the U.S. tech sector, focusing on entry- to mid-level roles that can plausibly be performed either remotely or in-person. Job postings will be systematically identified through major online job boards such as LinkedIn and Indeed, using consistent search criteria to ensure comparability across applications. For each job posting, resumes will be randomly assigned a gender signal and submitted accordingly. Employer responses will be tracked and recorded over a fixed period following submission. The dataset will be constructed at the application level, with each observation representing a single resume submission. Key variables will include the

callback indicator, gender assignment, job type (remote or in-person), and relevant identifiers such as job category or platform. The binary callback outcome may be supplemented with additional measures, such as response timing or type of response, where feasible.

Our primary hypothesis is that the gender gap in callback rates differs across work environments. To test this, we estimate the following regression model:

$$Callback_i = \alpha + \beta_1 Female_i + \beta_2 Remote_i + \beta_3 (Female_i \times Remote_i) + \varepsilon_i$$

where "callback" is a binary indicator equal to one if the application receives a positive employer response and zero otherwise. Female indicates whether the applicant has a female-signaling name, and remote indicates whether the job is remote rather than in-person. The interaction term female x remote captures whether the gender gap in callback rates differs between remote and in-person jobs.

The coefficient B1 measures the gender gap in callback rates for in-person jobs, while B3 captures how this gap changes in remote jobs relative to in-person jobs. We anticipate B1 to be negative, signaling the disparity in callback rates between male and female applicants. Our main parameter of interest is B3. Under the null hypothesis, B3=0, implying that the gender gap in callbacks is the same across both work environments. A positive B3 would indicate that the gender gap is smaller in remote jobs, while a negative B3 would indicate that the gap is larger in remote jobs.

Power Calculations

4.1 Setup and Parameter Assumptions

To determine the required sample size for our audit study, we conduct power calculations based on the structure of our experimental design. Our primary outcome is a binary callback indicator, and our main parameter of interest is the interaction term β_3, which captures the differential gender gap in callback rates between remote and in-person job postings. Because β_3 is an interaction term rather than a simple difference in means, detecting a meaningful effect requires greater statistical power than a standard two-sample comparison—and thus a larger sample.

We specify the following baseline parameters for our power calculations:

Significance level (α): We set $\alpha = 0.05$, corresponding to a 95% confidence level, which is standard in audit studies and experimental economics literature.

Statistical power (1 − β): We target a power of 0.80, meaning we require an 80% probability of detecting a true effect if one exists. We conduct sensitivity analysis at higher power levels (0.85

and 0.90).

Baseline callback rate: Drawing on prior audit studies in professional and technical occupations, we assume a baseline callback rate of approximately 20–25% for male applicants in in-person positions. This estimate is consistent with findings from comparable correspondence studies, such as Bertrand and Mullainathan (2004), who document callback rates in a similar range for favored applicant groups in professional settings. We treat $p_0 = 0.22$ as our baseline in the main calculation and vary it in our sensitivity analysis.

Minimum detectable effect (MDE): For the simple gender gap (β_1), we define the MDE as a difference of 5 percentage points in callback rates between male and female applicants in in-person jobs—$p_1 = 0.17$ for women. This is consistent with effect sizes documented in experimental studies of gender discrimination in STEM hiring contexts (Moss-Racusin et al. 2012), though we note that the precise MDE will be calibrated following our pilot data collection phase. For the interaction term β_3, which is our primary coefficient of interest, detecting a differential effect of the same magnitude requires a larger effective sample.

Clustering: Each job posting receives one male-signaling and one female-signaling resume. Because two resumes submitted to the same posting may yield correlated outcomes—for instance, if an employer has a uniform policy regarding a given position—we account for within-posting correlation and plan to cluster standard errors at the job posting level in our main specification.

4.2 Sample Size Estimation

Given a two-tailed test, a baseline callback rate of $p_0 = 0.22$ for male applicants, and an MDE of 5 percentage points ($p_1 = 0.17$ for female applicants), the required number of observations per group can be approximated using the standard formula for the difference in two proportions:

$$n = (z\alpha/2 + z\beta)^2[p0(1 - p0) + p1(1 - p1)]/(p0 - p\ )^2$$

Substituting $z_{0.025} = 1.96$ and $z_{0.20} = 0.84$, this yields approximately **n ≈ 800 applications per cell**, for a total of roughly **3,200 resume submissions** across the four cells of our 2×2 design (male/female × remote/in-person).

However, because our main parameter of interest is the interaction term β_3 rather than a simple difference in means, the power of our test is determined by our ability to detect the differential gender gap across work modalities. The variance of an interaction estimator is approximately four times that of a simple difference, meaning that the required sample size to achieve equivalent power is correspondingly larger. We therefore inflate our target sample by approximately 20–25%, bringing the total to roughly **3,800–4,000 applications**.

Additionally, to account for within-posting clustering, we apply a design effect correction. Assuming an intra-cluster correlation (ICC) of $\rho = 0.05$—a conservative assumption given that each cluster contains only two observations—the design effect is $1 + (2-1) \times 0.05 = 1.05$, a modest adjustment that does not substantially change our sample size requirements.

Finally, anticipating an application-level attrition rate of up to 15% due to job postings being removed before employer review, we inflate the total submission target to approximately **4,400–4,600 applications**. Submitted resumes where the posting is confirmed to have been removed prior to any employer engagement will be excluded from the analysis sample and documented separately.

4.3 Sensitivity Analysis

To assess the robustness of our sample size estimates, we are going to conduct two sensitivity analyses: one varying the MDE while holding power constant, and one varying the power level while holding the MDE constant. These analyses are intended to confirm that our target sample remains feasible across a range of plausible assumptions about baseline callback rates and effect sizes.

Figure 1 presents the required sample size per cell as a function of the MDE, holding $\alpha = 0.05$ and power = 0.80 fixed. As expected, the required sample size increases sharply as the MDE decreases toward zero. At our baseline MDE of 5 percentage points, the required per-cell sample is approximately 800 observations. Even under a more demanding MDE of 3 percentage points — which would correspond to a very small but policy-relevant differential effect — the required per-cell sample remains below 2,200, well within the range of feasibility given the volume of tech job postings in Virginia at any given time.

Figure 2 is going to present the required sample size per cell as a function of the power level, holding the MDE fixed at 5 percentage points and $\alpha = 0.05$. Moving from 80% to 90% power raises the required per-cell sample from approximately 800 to approximately 1,050, an increase of around 30%. Even at 90% power, the total required submissions — approximately 4,200 per cell after attrition adjustment, or roughly 5,800 in total — remain achievable over a 6–8 week data collection window given the density of tech job postings on LinkedIn and Indeed in the job market.

Taken together, these sensitivity analyses should be able to confirm that our study is feasible under a wide range of parameter assumptions. The results also highlight the value of a larger realized sample: if our data collection exceeds our minimum target, we will have sufficient power to detect smaller interaction effects, increasing the precision and policy relevance of our findings.

**Note: Figures 1 and 2 will be produced using Stata following the completion of a pilot data*

collection phase, in which we will submit a small number of applications to calibrate our baseline callback rate estimate prior to the full study.

Bibliography

Bertrand, Marianne, and Sendhil Mullainathan. (2004). "Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination." American Economic Review 94 (4): 991–1013.

Galos, D. R., & Coppock, A. (2023). Gender composition predicts gender bias: A meta-reanalysis of hiring discrimination audit experiments. Science Advances, 9(18). https://doi.org/10.1126/sciadv.ade7979

Goldin, C., & Rouse, C. (2000). Orchestrating Impartiality: The Impact of "Blind" Auditions on Female Musicians. American Economic Review, 90(4), 715-741. https://doi.org/10.1257/aer.90.4.715

Goldin, C. (2014). A Grand Gender Convergence: Its Last Chapter. American Economic Review, 104(4), 1091-1119. https://doi.org/10.1257/aer.104.4.1091

Moss-Racusin, C.A., Dovidio, J.F., Brescoll, V.L., Graham, M.J., & Handelsman, J. (2012) Science Faculty's Subtle Gender Biases Favor Male Students. Proc. Natl. Acad. Sci. U.S.A. 109 (41) 16474-16479. https://doi.org/10.1073/pnas.1211286109

Neumark, D. (2018). Experimental Research on Labor Market Discrimination. Journal of Economic Literature, 56(3), 799-866. https://doi.org/10.1257/jel.20161309

Robinson, B. (2026). "Tech Jobs Are Surging In These 10 States If You're Looking For One." Forbes.https://www.forbes.com/sites/bryanrobinson/2026/03/04/tech-jobs-are-surging-in-these-10-states-if-youre-looking-for-one/

U.S. Equal Employment Opportunity Commission. (2024). EEOC Research Finds Unequal Opportunity in the High Tech Sector and Workforce. https://www.eeoc.gov/newsroom/eeoc-research-finds-unequal-opportunity-high-tech-sector-and-workforce

WomenTech Network. (2025). "Women in Tech Stats 2026." https://www.womentech.net/women-in-tech-stats

Wulff Pabilonia, S., & Redmond, J. (2024). "The Rise in Remote Work since the Pandemic and Its Impact on Productivity : Beyond the Numbers." U.S. Bureau of Labor Statistics.https://www.bls.gov/opub/btn/volume-13/remote-work-productivity.htm#edn1